Combining Multi-Scale Context Aggregation, Attention-Guided Upsampling, and High-Resolution Feature Enhancement for Small Object Detection with YOLO

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***Abstract*—Small object detection remains one of the hardest open problems for single-stage detectors such as the YOLO family, because repeated stride-2 downsampling in the backbone erodes the spatial signal of small instances well before the detection heads can exploit it. In this work we study three complementary architectural interventions on YOLO11/YOLO12: (i) an attention-augmented, cross-stage-partial variant of the Spatial Pyramid Pooling – Fast (SPPF) block at the lowest-resolution stage (P5), combining a dilated multi-branch context path with Convolutional Block Attention Module (CBAM) refinement; (ii) a high-resolution early-stage branch that inserts lightweight attention modules at the P2/P3 stages and adds a dedicated P2 detection head, so that small-object evidence is exploited before it is downsampled away; and (iii) content-aware feature reassembly (CARAFE) in place of nearest-neighbor upsampling in the top-down feature pyramid, so that the detail reconstructed during top-down fusion is learned rather than interpolated. We evaluate each intervention in isolation and in combination against YOLO11, YOLO12, and a non-attention SPPF replacement, on two aerial/UAV small-object benchmarks, DOTA v1.5 and VisDrone. [TODO: One-sentence quantitative summary of the main result, e.g. "Our combined architecture improves mAP50-95 on small-object classes by X.X points over the YOLO11 baseline while adding Y.YM parameters."]**

***Index Terms*—small object detection, YOLO, spatial pyramid pooling, attention mechanism, feature pyramid network, aerial imagery, UAV imagery**

I. Introduction

Single-stage detectors of the YOLO family [1] have become the default choice for real-time object detection because they offer a favorable accuracy/latency trade-off. However, their performance on *small* objects — a few pixels wide after the backbone's successive stride-2 downsampling stages — lags substantially behind their performance on medium and large instances. This gap is particularly acute in aerial and UAV imagery, where the camera's bird's-eye viewpoint and high altitude make most instances of interest (vehicles, boats, pedestrians) occupy only a handful of pixels, and where datasets such as DOTA [9] and VisDrone [10] report that the majority of annotated instances fall under commonly used small-object area thresholds.

The root cause is architectural: by the time a feature map reaches the detector's deepest stage (P5, stride 32), a small object's spatial footprint has often collapsed to a single feature-map cell or less, and the information is further diluted by whatever aggregation happens at that stage — in YOLO11/YOLO12, the Spatial Pyramid Pooling – Fast (SPPF) [2] block, a cascade of same-kernel max-pooling operations. SPPF enlarges the effective receptive field cheaply, but it does so purely through max-pooling, which discards the fine-grained detail a small object needs and treats every channel and spatial location identically.

We argue — and structure our ablation study around the claim — that no single intervention at a single stage of the network is sufficient: a richer SPPF replacement at P5 only reshapes the *global context* that is later fused back into higher-resolution stages through the feature pyramid; it cannot recover detail already destroyed by downsampling. Conversely, an early-stage, high-resolution intervention that preserves small-object detail has no mechanism to inject the broader scene context that disambiguates visually similar small classes. We therefore propose and evaluate three interventions that act at three different points of the network, and measure whether their effects are complementary rather than redundant:

- **SPPFCSPC-CBAM** (Section III-B): a cross-stage-partial, attention-refined replacement for SPPF at P5, combining a CSP shortcut branch, a dilated multi-branch pooling cascade, and CBAM [6] channel/spatial attention.
- **TOSA + P2 head** (Section III-C): lightweight attention modules inserted at the P2 and P3 stages of the backbone, a soft (average-pooling-based) downsampling operator in place of strided convolutions between P2 and P4, and an additional detection head at stride 4 (P2), so that small-object evidence is used directly rather than only through a downsampled proxy.
- **CARAFE upsampling** (Section III-D): a content-aware reassembly operator [8] that replaces nearest-neighbor upsampling in the top-down feature pyramid path, so that the detail injected during top-down fusion is predicted

from local content rather than interpolated.

Our contributions are:

1) A systematic, controlled ablation of SPPF replacements for small object detection, isolating the respective contributions of multi-branch dilated context, CSP structure, and channel/spatial attention.
2) A combined architecture that integrates P5 context aggregation, P2/P3 high-resolution attention, and content-aware FPN upsampling into a single YOLO11-based model, evaluated end-to-end against each component in isolation.
3) An empirical comparison on two aerial/UAV benchmarks (DOTA v1.5, VisDrone) against YOLO11, YOLO12, and a non-attention SPPF replacement baseline. **[TODO: Confirm final experimental scope once all runs complete.]**

II. Related Work

A. YOLO and Spatial Pyramid Pooling

The YOLO family [1] evolved from YOLOv1's single dense prediction grid to multi-scale feature pyramids with a Spatial Pyramid Pooling (SPP) stage [2] at the backbone's deepest level, which YOLOv5 replaced with the cheaper, functionally equivalent SPPF (a single pooling kernel applied repeatedly instead of three parallel kernels of increasing size). YOLOv7 proposed SPPCSPC [3], which adds a cross-stage-partial shortcut around the pooling cascade, reducing the information bottleneck a plain SPP/SPPF introduces; our SPPFCSPC-CBAM module (Section III-B) extends this line of work with explicit multi-scale dilation and attention-based refinement.

B. Multi-scale context and receptive field diversity

Beyond SPP-style pooling, atrous/dilated convolutions have been used to enlarge the receptive field while preserving spatial resolution, notably in ASPP [5] for semantic segmentation and RFBNet's Receptive Field Block [4] for object detection, which fuses parallel dilated branches explicitly to diversify the effective receptive field rather than relying solely on pooling. Our SPPFCSPC-CBAM module borrows this multi-dilation-branch principle while retaining the cheap cascaded-pooling branch of SPPF for a controlled comparison.

C. Attention mechanisms

Channel and spatial attention modules have been shown to cheaply improve feature discriminability without significantly increasing model size. CBAM [6] sequentially applies channel and spatial attention; SimAM [7] achieves a comparable effect with a parameter-free energy-based formulation. We use CBAM inside our SPPF replacement and note SimAM as a lighter-weight alternative evaluated as a secondary module in our codebase but outside the scope of the present ablation.

D. Small object detection

Dedicated small-object detection work generally falls into two families: architectural interventions that preserve or restore high-resolution detail (additional high-resolution detection heads, e.g. a P2/stride-4 head as used by several YOLO variants and by QueryDet-style cascades; and feature pyramid upsampling improvements such as CARAFE [8], which predicts content-aware reassembly kernels instead of using fixed nearest-neighbor or bilinear interpolation in the top-down path) and data/training interventions (tiled inference such as SAHI, copy-paste augmentation of small instances, and loss functions less sensitive to small-box IoU volatility such as Wise-IoU/SIoU variants). The present work focuses on the architectural family, and treats the P2-head and CARAFE directions as complementary, testing them jointly with our SPPF-stage contribution rather than in isolation.

E. Aerial and UAV object detection benchmarks

DOTA [9] and VisDrone [10] are the two benchmarks used in this study. Both are dominated by small, dense instances captured from an aerial or near-aerial viewpoint (vehicles, ships, pedestrians), which makes them a natural testbed for small-object architectural interventions, as opposed to ground-level benchmarks such as COCO where the small-object subset is a much smaller fraction of the data.

III. Proposed Method

[TODO: Insert block diagram of the combined architecture: backbone with TOSA/P2 head (Sec. III-C), SPPFCSPC-CBAM at P5 (Sec. III-B), and CARAFE upsampling in the FPN (Sec. III-D).]

Fig. 1. Overview of the proposed architecture.

All variants studied in this paper share the YOLO11 backbone/head topology (Fig. 1), and differ only in the modules described below, which can be composed independently. This lets us attribute any accuracy change to a specific, isolated architectural change rather than to a confound of several simultaneous modifications.

A. Baseline

The YOLO11 baseline backbone downsamples the input through five stages (P1–P5, strides 2–32) using strided 3×3 convolutions and C3k2 blocks, aggregates context at P5 with SPPF, refines it with a C2PSA attention block, and feeds a standard PANet-style feature pyramid (nearest- neighbor upsampling, concatenation, C3k2 fusion) into three detection heads at P3/P4/P5. YOLO12 is evaluated as a second, independent baseline.

B. SPPFCSPC-CBAM: attention-refined SPPF replacement

Plain SPPF aggregates context purely through a cascade of same-kernel max-pooling operations, which (i) discards fine-grained detail at every pooling step and (ii) treats all channels and spatial locations uniformly. Our replacement, SPPFCSPC-CBAM, addresses both limitations while keeping the parameter/FLOP overhead at P5 — the network's lowest-resolution, cheapest-to-widen stage — deliberately contained.

Given an input feature map $X \in \mathbb{R}^{C \times H \times W}$, the module splits computation into a cross-stage-partial shortcut branch and a pooling branch (Eq. 1):

$$X_1 = \mathrm{Conv}_{1\times1}(X), \quad X_2 = \mathrm{Conv}_{1\times1}(X) \tag{1}$$

$$X_1' = \mathrm{Conv}_{1\times1}(\mathrm{Conv}_{3\times3}(X_1)) \tag{2}$$

where X_2 bypasses pooling entirely and is fused back in at the end. The pooled branch applies a $k \times k$ max-pool n times in sequence (as in SPPF, equivalent to a single pass of $k, 2k, \ldots$ receptive fields at a fraction of the cost of parallel kernels):

$$Y_0 = X_1', \quad Y_i = \mathrm{MaxPool}_k(Y_{i-1}),\ i = 1, \ldots, n \tag{3}$$

The $n + 1$ pooled feature maps are concatenated, fused, and refined with CBAM channel-then-spatial attention [6] before being concatenated with the untouched shortcut branch and projected to the output channel count:

$$Y = \mathrm{CBAM}\big(\mathrm{Conv}_{3\times3}(\mathrm{Conv}_{1\times1}([Y_0, \ldots, Y_n]))\big) \tag{4}$$

$$\mathrm{Out} = \mathrm{Conv}_{1\times1}([Y, X_2]) \tag{5}$$

The CSP shortcut (X_2) guarantees that at least one path through the module carries the input forward with no lossy pooling, and CBAM re-weights the pooled branch by learned channel and spatial relevance rather than taking the pooling cascade's output at face value.

C. TOSA and the P2 detection head

SPPFCSPC-CBAM operates at P5 (stride 32) and therefore cannot recover detail already lost to downsampling; it only improves the quality of the context subsequently fused back into higher-resolution stages. To act *before* that detail is lost, we insert a lightweight attention module (TOSA) directly after the P2 (stride 4) and P3 (stride 8) stages of the backbone, replace the strided 3×3 convolutions that downsample P2→P3 and P3→P4 with a channel-preserving average-pooling depth-wise convolution (so that width scaling propagates through the backbone unchanged), and add a fourth detection head at P2/stride 4 (160×160 at a 640×640 input), alongside the standard P3/P4/P5 heads. The P2 head lets the detector directly supervise and predict from the highest-resolution feature map the network produces, rather than relying exclusively on heads that only see small objects after at least one more round of downsampling than P2 itself.

D. Content-aware upsampling (CARAFE)

The top-down path of a standard YOLO feature pyramid upsamples low-resolution, high-level features with fixed nearest-neighbor interpolation before fusing them with the corresponding higher-resolution backbone feature map. This interpolation is content-agnostic: it cannot reconstruct any detail beyond what the low-resolution map already encodes explicitly. We replace nearest-neighbor upsampling with CARAFE [8], which predicts, from the local content of the low-resolution feature map, a reassembly kernel used to redistribute each output pixel's value from a local neighborhood of input pixels, instead of simply repeating the nearest input value. This is channel-preserving and computationally lightweight (a small encoder plus an unfold-based weighted sum) relative to the convolutional blocks that already dominate the pyramid's cost.

E. Combined architecture

Our proposed architecture (denoted COMBO in Section V) applies all three interventions jointly on top of the YOLO11 topology: TOSA and the P2 head at the backbone's high-resolution stages, SPPFCSPC-CBAM at P5, and CARAFE throughout the top-down FPN path. Because each intervention acts at a different point of the network and is independently toggleable in our implementation, Section V reports both the combined model and each component in isolation, which lets us test directly whether their effects are additive, redundant, or interacting.

IV. Experimental Setup

A. Datasets

We evaluate on two aerial/UAV object detection benchmarks dominated by small, dense instances:

- **DOTA v1.5** [9]: 1,411 training and 458 validation images (the official `test` split, 937 images, has no public ground truth and is excluded from this study), 16 categories, annotated with oriented bounding boxes, converted to axis-aligned boxes for the detection task studied here.
- **VisDrone** [10]: drone-captured imagery across 10 object categories (pedestrian, vehicle, bicycle, etc.), characterized by dense small-object scenes with substantial occlusion.

B. Models compared

- **YOLO11 / YOLO12**: unmodified baselines.
- **GL**: SPPF replaced by a non-attention GL-CAB block at P5 only, isolating the effect of the architectural change itself from attention.
- **GLCCP / CCP**: two further SPPF-slot variants built on progressive dilated context aggregation (GL-PSWCA).

- **SCC**: our SPPFCSPC-CBAM module (Section III-B) in isolation, all other stages unchanged from YOLO11.
- **T**: TOSA + P2 head (Section III-C) in isolation, SPPF unchanged.
- **COMBO**: our full proposed architecture (Section III-E).

C. Training protocol

All models are trained from the corresponding YOLO11n/YOLO12n COCO-pretrained checkpoint, for **[TODO: 200]** epochs at **[TODO: 1280]**px, batch size 16 (**[TODO: except COMBO, trained at batch 8 on VisDrone due to a GPU memory constraint – confirm which runs this applied to]**), with SGD (lr0$=0.01$, lrf$=0.01$, momentum$=0.937$, weight decay$=5\times10^{-4}$), a cosine learning-rate schedule, and the standard YOLO detection loss (box$=7.5$, cls$=0.5$, dfl$=1.5$). All variants are trained with identical hyperparameters to isolate the effect of the architectural change from any optimization confound, with the batch-size exception above noted as a deviation.

a) Weight initialization.: Pretrained weights are transferred tensor-by-tensor, matched by exact parameter name and shape between the COCO checkpoint and the target architecture; any tensor that does not match keeps PyTorch's default (random) initialization. Because layers are named by their sequential position in the network, how much of a variant actually inherits COCO-pretrained weights depends on *where* its architectural changes sit, not merely on whether `pretrained=True` was set. We measured the exact transferred/total tensor count for every variant (Table I): single-layer, same-position replacements (GL, SCC) leave every other layer's naming and shape unchanged, so >92% of tensors still transfer; variants that insert additional layers early in the backbone (T, COMBO) or replace a block type network-wide (GLCCP, CCP) shift or rename most subsequent layers, so only ≈8–10% of tensors transfer – in practice close to training from scratch beyond the first stem convolution(s), despite `pretrained=True`. We report this because it is a potential confound for the comparison: part of an observed accuracy gap could stem from this initialization difference rather than from the architectural change under study. **[TODO: Decide whether to control for this (e.g. report both pretrained and from-scratch runs for the heavily-reindexed variants) or report it as a limitation.]**

TABLE I
FRACTION OF MODEL TENSORS INITIALIZED FROM THE COCO-PRETRAINED CHECKPOINT (VS. LEFT AT PYTORCH DEFAULT INIT), SCALE N

Variant	Transferred	Total	%
YOLO11 (baseline)	499	499	100.0
GL	487	511	95.3
SCC (ours)	493	532	92.7
GLCCP	161	1749	9.2
CCP	173	1745	9.9
T (ours)	56	611	9.2
COMBO (ours)	56	665	8.4

D. Evaluation metrics

We report precision (P), recall (R), mAP_{50}, and $mAP_{50\text{-}95}$, computed on the validation split with a confidence threshold of 0.001 and an NMS IoU threshold of 0.7, matching the YOLO validation protocol so that results are directly comparable across variants. **[TODO: Add a small-object-only breakdown (e.g. COCO-style area$< 32^2$px bucket) once per-size metrics are available, since aggregate mAP can dilute a small-object-targeted improvement.]**

V. RESULTS

[TODO: All tables below are placeholders pending completion of full-length (200-epoch) training runs for every model/dataset pair. Do not submit with placeholder values.]

TABLE II
DOTA V1.5 – VALIDATION RESULTS

Model	P	R	mAP_{50}	$mAP_{50\text{-}95}$
YOLO11	[TODO:]	[TODO:]	[TODO:]	[TODO:]
YOLO12	[TODO:]	[TODO:]	[TODO:]	[TODO:]
GL	[TODO:]	[TODO:]	[TODO:]	[TODO:]
GLCCP	[TODO:]	[TODO:]	[TODO:]	[TODO:]
CCP	[TODO:]	[TODO:]	[TODO:]	[TODO:]
SCC (ours)	[TODO:]	[TODO:]	[TODO:]	[TODO:]
T (ours)	[TODO:]	[TODO:]	[TODO:]	[TODO:]
COMBO (ours)	[TODO:]	[TODO:]	[TODO:]	[TODO:]

TABLE III
VISDRONE – VALIDATION RESULTS

Model	P	R	mAP_{50}	$mAP_{50\text{-}95}$
YOLO11	[TODO:]	[TODO:]	[TODO:]	[TODO:]
YOLO12	[TODO:]	[TODO:]	[TODO:]	[TODO:]
GL	[TODO:]	[TODO:]	[TODO:]	[TODO:]
GLCCP	[TODO:]	[TODO:]	[TODO:]	[TODO:]
CCP	[TODO:]	[TODO:]	[TODO:]	[TODO:]
SCC (ours)	[TODO:]	[TODO:]	[TODO:]	[TODO:]
T (ours)	[TODO:]	[TODO:]	[TODO:]	[TODO:]
COMBO (ours)	[TODO:]	[TODO:]	[TODO:]	[TODO:]

TABLE IV
MODEL SIZE AND COMPUTE COST (SCALE N)

Model	Params (M)	GFLOPs
YOLO11	2.59	[TODO:]
GLCCP (CCP)	2.79	[TODO:]
SCC (ours)	4.27	[TODO:]
T (ours)	2.49	[TODO:]
COMBO (ours)	4.37	[TODO:]

A. Ablation discussion

[TODO: Once Tables II–III are populated, discuss: (i) whether SCC alone improves over GL/CCP, isolating the contribution of the CSP shortcut and CBAM attention from the dilation-based context aggregation used by GL/CCP; (ii) whether T alone (early-stage, P2 head)

outperforms every P5-only variant, as argued in Section I on resolution grounds; (iii) whether COMBO's improvement over the best single-intervention variant is closer to additive or to redundant, which is the central empirical question this ablation is designed to answer.]

VI. Conclusion

We presented a controlled ablation of three complementary interventions for small-object detection in YOLO11/YOLO12 – an attention-refined SPPF replacement at the network's lowest-resolution stage, a high-resolution early-stage attention branch with a dedicated P2 detection head, and content-aware FPN upsampling – and a combined architecture that integrates all three. **[TODO: One or two sentences summarizing the final quantitative conclusion once Section V is complete.]** Future work includes extending the ablation to inference-time techniques orthogonal to architecture (tiled/sliced inference, small-object-aware NMS, and IoU-loss variants less sensitive to small-box localization error), and evaluating transfer to additional aerial benchmarks beyond DOTA and VisDrone.

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